Since $\mathbb{F}_{q^2}^\times$ is cyclic and $\nu^q \neq \nu$,

$$\sum_{\zeta \in \mathbb{F}_{q^2}^\times} \nu^{q-1}(\zeta) = \sum_{\zeta \in \mathbb{F}_{q^2}^\times} \nu^{1-q}(\zeta) = 0.$$

Therefore,

$$\sum_{\zeta \text{ elliptic}} (\nu^{q-1}(\zeta) + \nu^{1-q}(\zeta)) = -\sum_{\zeta \in \mathbb{F}_q^\times} (\nu^{q-1}(\zeta) + \nu^{1-q}(\zeta)) = -2(q-1)$$

since $\mathbb{F}_q^\times$ is cyclic of order $q-1$. Therefore,

$$\begin{aligned}\langle \chi, \chi \rangle = \frac{1}{(q-1)^2 q(q+1)} \big(&(q-1)\cdot(q-1)^2 \cdot 1 + (q-1)\cdot 1 \cdot (q^2-1) \\ &+ \frac{q(q-1)}{2} \cdot (2(q^2-q) - 2(q-1)) \big) = 1.\end{aligned}$$

□

We have now shown that for any ν with $\nu^q \neq \nu$ the representation Y_ν with the same character as

$$W_1 \otimes V_{\alpha,1} - V_{\alpha,1} - \operatorname{Ind}_K^G \mathbb{C}_\nu$$

exists and is irreducible. These characters are distinct for distinct pairs $(\alpha,\ \nu)$ (up to switching $\nu \to \nu^q$), so there are $\frac{q(q-1)}{2}$ such representations, each of dimension $q-1$. These representations are called complementary series representations.

We have thus found $q-1$ 1-dimensional representations of G, $\frac{q(q-1)}{2}$ principal series representations, and $\frac{q(q-1)}{2}$ complementary series representations, for a total of q^2-1 representations, i.e., the number of conjugacy classes in G. This implies that we have in fact found all irreducible representations of $GL_2(\mathbb{F}_q)$.

5.26. Artin's theorem

Theorem 5.26.1. *Let X be a conjugation-invariant system of subgroups of a finite group G. Then two conditions are equivalent:*

(i) Any element of G belongs to a subgroup $H \in X$.

(ii) The character of any irreducible representation of G belongs to the $\mathbb{Q}$-span of characters of induced representations $\operatorname{Ind}_H^G V$, where $H \in X$ and V is an irreducible representation of H.